Concept Flyer — Shunyaya Structural Computation Core (SSC-Core)

Deterministic Structural Work Geometry & Execution Topology Framework

---

Version: Public Open Standard (v1.4)
Release Date: March 5, 2026
Status: Public Open Standard (Deterministic Structural Computation Layer)
License: Open Standard — specification may be implemented freely; provided "as is" without warranty or liability.
Caution: Deterministic structural work measurement and replay verification only.

The Problem — Computation Produces Output but Not Geometry

Classical systems evaluate:

- Is the output correct?
- Is the equation satisfied?
- Does the program terminate?

They do not measure:

- How work was distributed
- Whether structural compression occurred
- Whether execution oscillated
- Whether identical outputs were produced by identical topology

Two programs can produce identical `m_out`.

Classical computation cannot determine whether their execution geometry was identical.

There is no deterministic vocabulary for structural work topology.

The Shift — From Output Evaluation to Execution Geometry

SSC-Core does not modify classical computation.
SSC-Core does not predict or optimize.

It introduces a conservative deterministic extension:

Each execution becomes measurable as structural geometry.

Core quantities:

- Structural work field `Psi(t)`
- Compression integral `Omega`
- Structural curvature `K`
- Structural density `D`
- Deterministic topology fingerprint `F`

Classical output remains unchanged.
Structural geometry becomes computable.

Core Invariant — Output Is Never Altered

Non-negotiable collapse invariant:

```
phi((m,a,s)) = m
```

Where:

- `m` = classical magnitude
- `a` = structural alignment
- `s` = structural posture

SSC-Core overlays structural geometry only.
SSC-Core never alters output or domain logic.

What SSC-Core Computes

Structural work field:

```
Psi(t) = W_allow(t) / W(t)
```

Structural compression:

```
Omega = sum_{i=1..N} (1 - Psi(i))
```

Structural curvature:

```
K = sum_{i=2..N} |g(i) - g(i-1)|
```

Structural density:

```
D = W_allow(N) / W(N)
```

Topology fingerprint:

```
F = hash(g_stream || "|" || Psi_samples || "|" || Omega || "|" || K)
```

All quantities are deterministic, trace-derived, and replay-stable.
They are independent of output magnitude.

SSC-Core measures geometry of work — not value semantics.

Replay Authority — Structural Identity Is Verifiable

SSC-Core enforces strict replay identity:

```
B_A = B_B
```

Structural identity requires:

- Byte-identical trace artifacts
- Identical structural quantities
- Identical topology fingerprint `F`
- Identical manifest hash

No tolerance.
No approximate equality.

Identity is strictly computational.

Structural Divergence Becomes Measurable

Without structural geometry:

- Identical outputs mask topology differences
- Compression instability remains invisible

SSC-Core makes execution topology explicit.
Structural divergence becomes explicitly measurable.

Deterministic Discipline

Conforming runs require:

- Declared deterministic work model
- Deterministic decision stream `g(i)`
- Trace-derived recomputation
- Manifest enforcement
- Tamper ritual validation

All runs must satisfy:

```
phi((m,a,s)) = m
B_A = B_B
```

Determinism is demonstrated — not asserted.

Why SSC-Core Matters

- Adds structural geometry to computation
- Enables topology comparison across implementations
- Preserves classical correctness
- Provides replay-verifiable identity certification

SSC-Core extends computation
from output-only evaluation
to measurable execution topology.

What SSC-Core Is Not

SSC-Core does not:

- Predict performance
- Replace algorithm analysis
- Compute entropy
- Estimate probability
- Prove correctness

It measures deterministic execution geometry only.

Architectural Position

Classical Layer → Output `m_out`
Work Model → `W(t)`
Structural Field → `Psi(t)`
Compression → `Omega`
Curvature → `K`
Density → `D`
Topology Identity → `F`

All collapse through:

```
phi((m,a,s)) = m
```

Magnitude remains primary.
Structure becomes measurable.

Closing Principle

Classical systems ask:

“What is the output?”

SSC-Core asks in parallel:

“What is the measurable geometry of the work that produced it?”

When execution becomes structurally measurable and replay-verifiable,
identity extends beyond value.

Without modifying output.

OMP